

Hariprasath Parthasarathy

Senior Software Engineer | Vancouver, BC | US: H-1B | Canada: Permanent Resident
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SUMMARY

Software engineer with 7+ years specializing in database internals in C and C++, including 3.5 years in Oracle's database kernel building Raft-based sharding replication and earlier storage-engine and JIT query-executor work on PostgreSQL systems. Strong fundamentals in data structures, algorithms, and software design.

EXPERIENCE

- **2by4.ai, Co-founder & CTO** San Francisco, CA
July 2025 – Present
AI pre-construction platform · Backed by Entrepreneurs First & Transpose Platform
 - Architected a multi-stage estimation pipeline combining LLMs, VLMs, and computer vision that converts hundreds of pages of construction blueprints into accurate material takeoffs, cutting estimate turnaround from 2 weeks to 3 days.
 - Scaled to 30 projects across 18 customers and 5 paid pilots within 6 weeks of launch, closing \$48K in annual recurring revenue (CARR), building a \$2.5M pipeline, and completing estimates on \$75M+ of construction work.
- **Vluna, Co-founder & CEO** San Francisco, CA
December 2024 – June 2025
Secure container image platform · Entrepreneurs First
 - Built a custom package manager and container build system from scratch that compiles every dependency from source, producing hardened, near-zero-CVE images (roughly 90% attack-surface reduction) with automated daily rebuilds and real-time patching, meeting security-mitigation SLAs at lower cost than Chainguard.
 - After discovery with 20+ CISOs and 40+ DevSecOps engineers converged the team on the \$42B container-security market, signed 3 design partners including a unicorn (Ironclad), with 50+ image pulls within 2 days of launch.
- **Oracle Corporation** Redwood City, CA
June 2021 – December 2024
Senior Database Kernel Engineer, Oracle Database
 - Owned the storage layer of Sharding Native Replication (23ai), the Raft-based engine that replaced Oracle's external replication tooling with active/active shards and automatic sub-second failover; built core components for shard coordination and leader/follower state management in C/C++.
 - Designed and implemented the IO and serialization layer that persists and ships Raft log records across replication units; tested and hardened recovery paths under node failure, network partition, and message loss.
 - As part of the team that deployed Oracle Database for Bank of America, processing \$2.5 trillion in transactions, contributed design reviews, code reviews, and production C/C++ across the distributed database stack.
- **Amazon** Seattle, WA
March 2020 – June 2021
Software Development Engineer, Amazon Devices
 - Redesigned and migrated Amazon Devices' externally facing returns microservice of 17 APIs (Java, Spring Boot, React), optimizing database queries and fetches to cut API latency by 20%.
- **Zoho Corporation** Chennai, India
May 2017 – May 2018
Member Technical Staff, Zoho Labs (Research and Development)
 - Extended CitusDB's CStore columnar FDW (C, LLVM) with in-memory caching, deletes, and a JIT query executor, improving query performance 8–10x on OLAP analytics queries over 30M rows.
 - Prototyped an in-memory PostgreSQL foreign data wrapper with a custom storage engine and query executor, using lock-free Bw-tree indexing and write-ahead logging for high-throughput workloads.

TECHNICAL SKILLS

- **Languages:** C++, C, Java, Python, JavaScript, SQL
- **Databases and Systems:** Query engines, query optimization, storage engines, columnar/OLAP storage, distributed databases, sharding, replication, high availability, Raft consensus, JIT query compilation, indexing, write-ahead logging, PostgreSQL, MySQL
- **Frameworks and Tools:** LLVM, Docker, Kubernetes, Kafka, Linux/LXC containers, Spring Boot, React, Git
- **AI and ML:** LLMs, VLMs, computer vision

PROJECTS AND AWARDS

- **Fastest Query Processor, CSE 562 (Database Systems):** Built a query processor that placed first for execution speed against the full graduate class.
- **Best MS Researcher, University at Buffalo:** Recognized for graduate research contributions in database and systems.

EDUCATION

- **University at Buffalo (State University of New York)** Buffalo, NY
August 2018 – December 2019
Master of Science in Computer Science and Engineering
- **Anna University** Chennai, India
2013 – 2017
Bachelor of Engineering in Computer Science and Engineering